

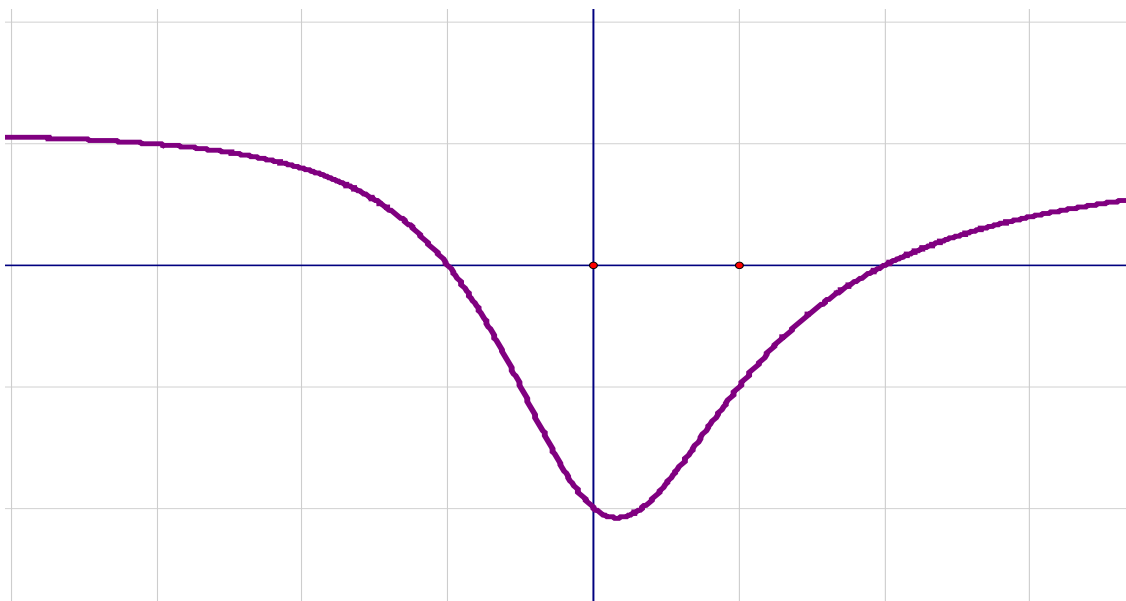
JRAHS Trial HSC 2006 Extension 2

Question 1 (15 Marks)	Marks
(a) Find $\int \frac{2x}{1+2x} dx$.	2
(b) How many different ways are there of choosing a cricket 11 from a team of 15 players, if there can only be at most, one of the 2 Lee brothers and 2 of 3 Abey brothers?	3
(c) Find $\lim_{x \rightarrow -5} \frac{\sqrt{20-x} - 5}{5+x}$.	2
(d) Find the sum of all the coefficients of powers of z in the expansion $(1+z)^8$.	2
(e) (i) Find the equation of the tangent to the hyperbola $3x^2 - y^2 = 12$ at the point $T(x_1, y_1)$.	3
(ii) This tangent meets the line $d(x=1)$ in the point M . Prove that $FM \perp FT$, where F is the focus $(4, 0)$.	3

Question 2 (15 Marks) START A NEW PAGE

(a) (i) Express $z = \sqrt{2} - i\sqrt{2}$ in modulus – argument form.	2
(ii) Hence, write z^{22} in the form of $a + ib$, where a and b are real.	3
(b) If a, b, c are real and unequal and that $a^2 + b^2 > 2ab$ deduce that	
(i) $a^2 + b^2 + c^2 > ab + bc + ca$	2
(ii) If $a + b + c = 6$ show that $ab + bc + ca < 12$	2
(c) Let $I_n = \frac{1}{n!} \int_0^1 x^n e^{-x} dx$ for $n \geq 0$.	
(i) Use integration by parts to show that $I_n = I_{n-1} - \frac{e^{-1}}{n!}$	3
(ii) Hence, evaluate I_4 .	3

- (a) The diagram below is a sketch of the function $y = f(x)$.



On separate diagrams sketch neatly:

- | | |
|--|---|
| (i) $y = f(x) $ | 2 |
| (ii) $y = e^{f(x)}$ | 2 |
| (iii) $y = \ln(f(x))$ | 2 |
| | |
| (b) Reduce the polynomial $x^4 - 2x^2 - 15$ over the rational and the complex field. | 2 |
| | |
| (c) If z_1, z_2 are 2 complex numbers such that $ z_1 + z_2 = z_1 - z_2 $.
With the aid of a diagram, show that $\arg(z_1)$ and $\arg(z_2)$ differ by either $\frac{\pi}{2}$ or $\frac{3\pi}{2}$. | 4 |
| | |
| (d) Evaluate $\int_0^{\frac{\pi}{2}} \frac{1 - \tan x}{1 + \tan x} dx$. | 3 |

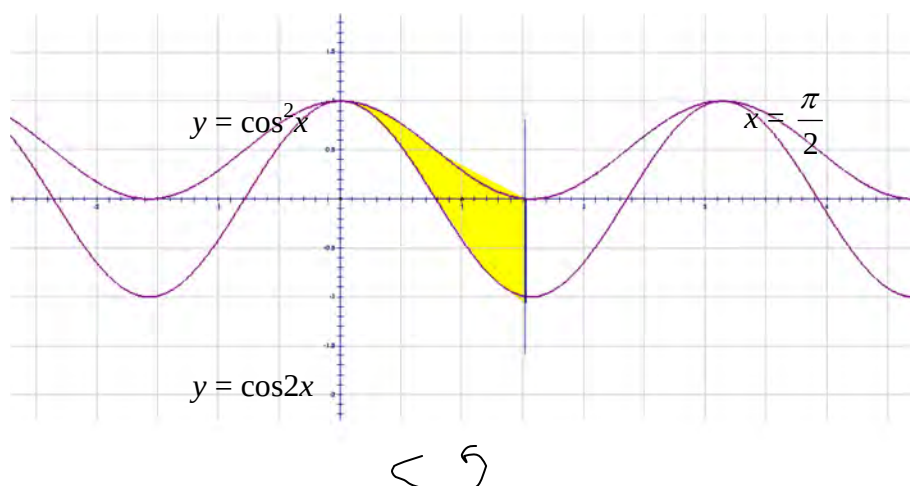
Question 4 (15 Marks)
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Marks

- (a) A string 50 cm of length can just sustain a weight of mass 20 kg without breaking. A mass of 4 kg is attached to one end of the string and revolves uniformly on a smooth horizontal table; the other end is being fixed to a point on the table. 3

Find the greatest number of complete revolutions the mass can make in a minute without breaking the string. [Use acceleration due to gravity as 9.8 m/s^2]

- (b) Evaluate $\int_a^{a^2} \frac{dx}{x \ln x}$, leaving your answer in simplified, exact form. 3

(c)



The area between the graphs of $y = \cos^2 x$ and $y = \cos 2x$ for $0 \leq x \leq \frac{\pi}{2}$ 6

and $x = \frac{\pi}{2}$ is rotated about the y -axis. By considering cylindrical shells, find volume of revolution of the solid formed, in terms of π .

- (d) Use the properties of odd and even functions to evaluate $\int_{-4}^4 \cos x (e^x - e^{-x}) dx$. 3

Question 5 (15 Marks) START A NEW PAGE

(a) (i) Find A and B for which $\frac{x^4}{x^2+1} = A(x^2-1) + \frac{B}{x^2+1}$. **2**

(ii) Show that this result may be used in the integration of the function $x^3 \tan^{-1} x$. **5**

Hence, evaluate the integral $\int_0^1 x^3 \tan^{-1} x \, dx$.

- (b) A particle P is projected vertically upwards from the surface of the Earth with initial speed u . The acceleration due to gravity at any point on its path is inversely proportional to the square of its distance from the centre of the Earth.

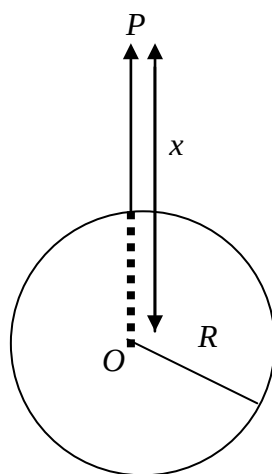


Diagram not to scale

(i) Prove that the speed v in any position x is given by $v^2 = u^2 - 2gR + \frac{2gR^2}{x}$, **4**

where R is the radius of the Earth and g is the acceleration due to gravity at the surface of the Earth.

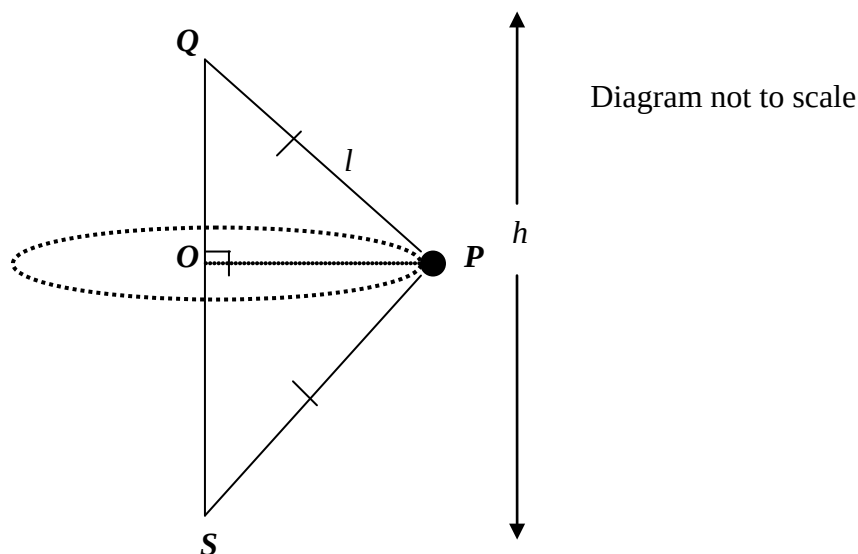
(ii) If $u = \sqrt{2gR}$, find the time taken to reach a height $3R$ above the surface of the earth, in terms of g and R . **4**

Question 6 (15 Marks)

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Marks

- (a) A total of five players is selected at random from four sporting teams. Each of the teams consists of 10 players numbered from 1 to 10. What is the probability that the five selected players contain at least four players from the same team? Clearly explain your answer. 4
- (b) A particle P of mass m is attached by two equal, light inextensible strings of length l , to two points Q and S , distant h units apart in the same vertical line; S is directly below Q . P rotates in a horizontal circle with uniform angular velocity, ω .



- (i) Prove that the tension in the string PQ is $ml\left(\frac{1}{2}\omega^2 + \frac{g}{h}\right)$ where g is the acceleration due to gravity. 5
- (ii) Find the tension in the string PS . 1
- (iii) Show that for both strings to remain stretched then $\omega > \sqrt{\frac{2g}{h}}$. 2
- (iv) If the tensions in the string are in the ratio of 2:1, find the period of the motion of the particle. 3

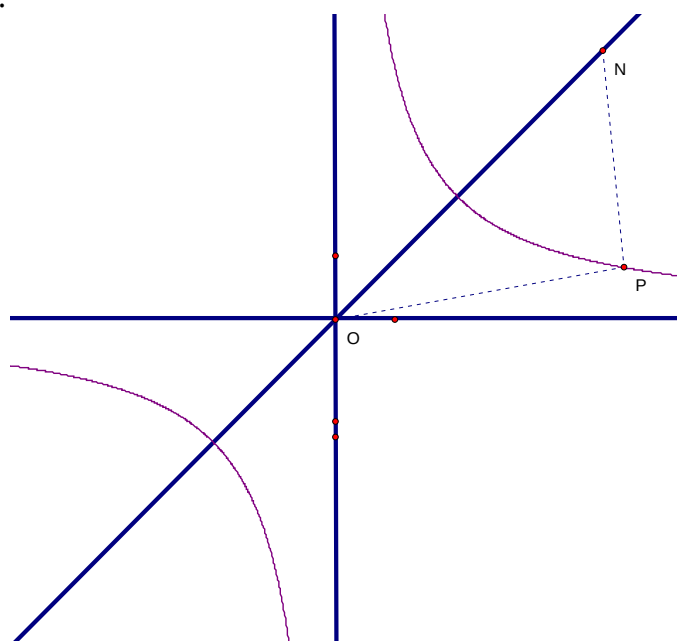
Question 7 (15 Marks)
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Marks

- (a) α, β, γ are non – zero roots of the equation $x^3 + px + q = 0$.
Find the equation whose roots are

(i) $\alpha^3, \beta^3, \gamma^3$ 3

(ii) $\frac{\alpha}{\beta\gamma}, \frac{\beta}{\alpha\gamma}, \frac{\gamma}{\alpha\beta}$ 4

- (b) The diagram below shows the hyperbola $xy = c^2$. The point $P\left(ct, \frac{c}{t}\right)$ lies on the curve, where $t \neq 0$. The normal at P intersects the straight line $y = x$ at N . O is the origin.



- (i) Prove that the equation of the normal at P is 1

$$y = t^2x + \frac{c}{t} - c^3.$$

- (ii) Find the coordinates of N . 1

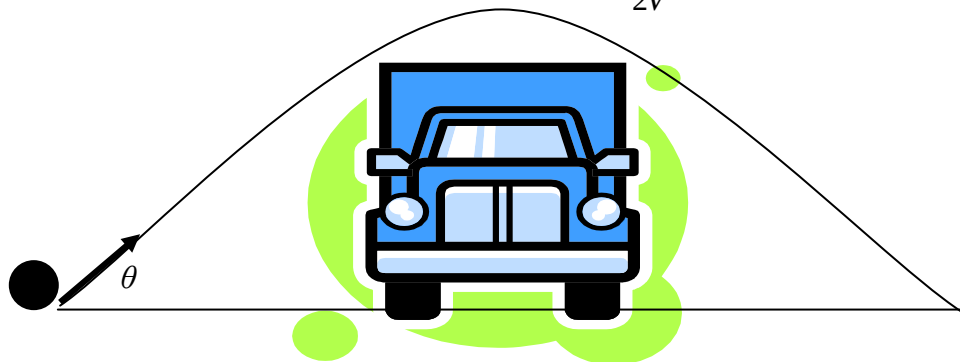
- (iii) Show that $\triangle OPN$ is isosceles. 2

- (c) (i) Draw a neat sketch of the graph of $y = \frac{4}{x} - x$. 2

- (ii) Draw a neat sketch of the curve $y = \sqrt{f(x)}$. 2

- (a) Ben's Hot Wheel travels along an inclined ramp to jump over a truck. The Hot Wheel travels at an initial speed V m/s inclined at an angle θ to the horizontal at O , and acceleration due to gravity is g m/s².

The Wheel's trajectory is given by $y = x \tan \theta - \frac{gx^2 \sec^2 \theta}{2V^2}$.



- (i) Write this equation in the general form of a parabola: $(x - h)^2 = 4a(y - k)$, where a, h, k are constants. 3
- (ii) Calculate the angle of projection, θ , if the range is three times the width of the truck and the top of the truck passes through the focus of equation in part (a). 3
- (b) Given p and q are positive real numbers,
- (i) Prove that: $\frac{1}{2}(p + q) \geq \sqrt{pq}$. 1
- (ii) Hence, deduce that: $\sqrt{p} \leq \frac{1}{2} \left(\frac{p}{\sqrt{q}} + \sqrt{q} \right)$. 1

Question 8 Part (c) continued on the next page

- (c) Given $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are positive real numbers, where $A_n = a_1 + a_2 + a_3 + \dots + a_n$ and $B_n = b_1 + b_2 + b_3 + \dots + b_n$, are such that $a_1, a_2, \dots, a_n > 0$, $b_1, b_2, \dots, b_n > 0$ and $A_r \leq B_r$, for $r = 1, 2, 3, \dots, n$.

- (i) Prove, by mathematical induction for $n = 1, 2, 3, \dots$, that: **3**

$$\begin{aligned} & \frac{1}{\sqrt{b_n}} B_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) B_{n-1} + \left(\frac{1}{\sqrt{b_{n-2}}} - \frac{1}{\sqrt{b_{n-1}}} \right) B_{n-2} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 \\ &= \sqrt{b_1} + \sqrt{b_2} + \sqrt{b_3} + \dots + \sqrt{b_n}. \end{aligned}$$

- (ii) Hence, given: $\frac{a_1}{\sqrt{b_1}} + \frac{a_2}{\sqrt{b_2}} + \frac{a_3}{\sqrt{b_3}} + \dots + \frac{a_n}{\sqrt{b_n}} =$

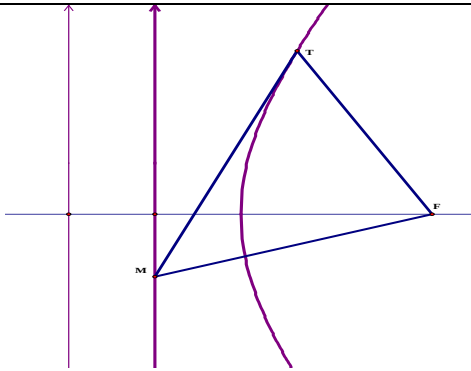
$$\frac{1}{\sqrt{b_n}} A_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) A_{n-1} + \left(\frac{1}{\sqrt{b_{n-2}}} - \frac{1}{\sqrt{b_{n-1}}} \right) A_{n-2} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) A_1,$$

show that: $\sum_{r=1}^n \frac{a_r}{\sqrt{b_r}} \leq \sum_{r=1}^n \sqrt{b_r}.$ **1**

- (iii) Deduce that: $\sum_{r=1}^n \sqrt{a_r} \leq \sum_{r=1}^n \sqrt{b_r}.$ **3**

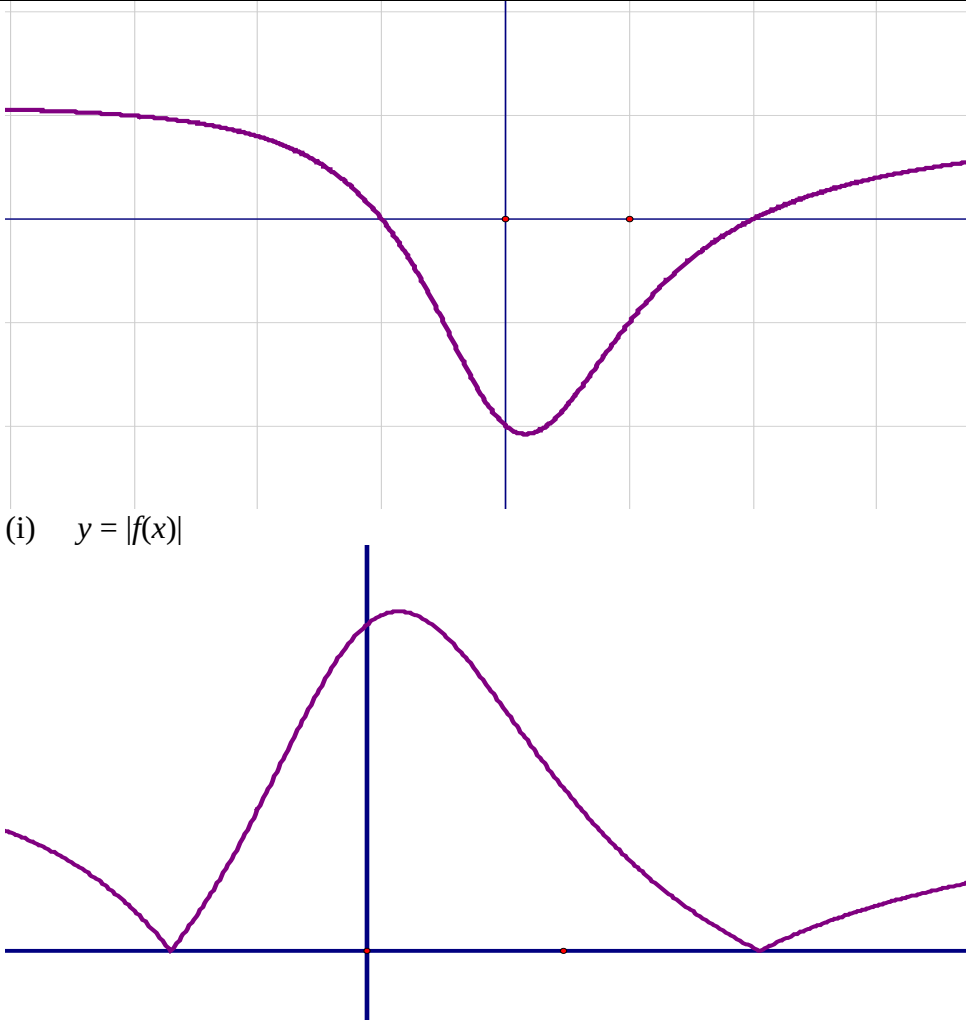
Year 12 Mathematics Extension 2 TRIALS 2006 – Suggested Solutions (LK)

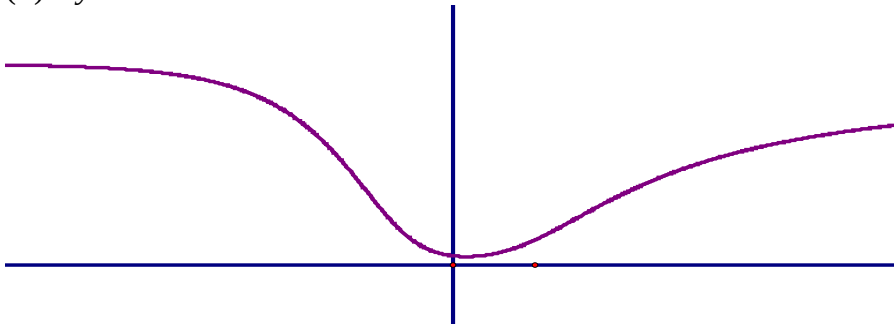
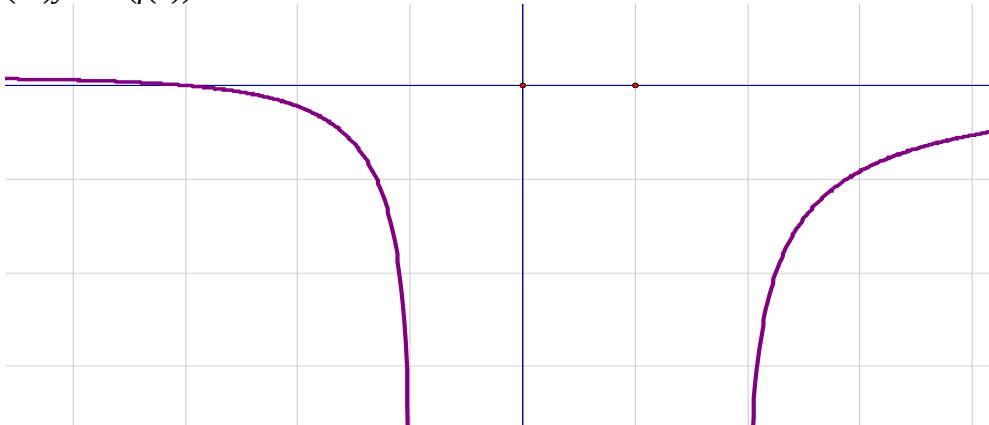
Solutions to Questions	Marking Scheme	Comments																		
Question 1																				
(a) $\int \frac{2x}{1+2x} dx = \int 1 - \frac{1}{1+2x} dx$ $= x - \frac{1}{2} \ln 1+2x + c$	1 for changing the integral 1 correct integration																			
(b) Let the Lee brothers be L_1 and L_2 Let the Abey brothers be A_1, A_2 and A_3. Then there are 10 others. $\therefore$ # possibilities is 365 ways. <table border="1"> <thead> <tr> <th>L</th><th>A</th><th>Others</th></tr> </thead> <tbody> <tr> <td>1</td><td>0</td><td>10</td></tr> <tr> <td>1</td><td>1</td><td>9</td></tr> <tr> <td>1</td><td>2</td><td>8</td></tr> <tr> <td>0</td><td>1</td><td>10</td></tr> <tr> <td>0</td><td>2</td><td>9</td></tr> </tbody> </table> $\therefore 2 + 60 + 270 + 3 + 30 = \mathbf{365}$ [using nC_r]	L	A	Others	1	0	10	1	1	9	1	2	8	0	1	10	0	2	9	1 for correct answer 2 for justification	
L	A	Others																		
1	0	10																		
1	1	9																		
1	2	8																		
0	1	10																		
0	2	9																		
(c) $\lim_{x \rightarrow -5} \frac{\sqrt{20-x}-5}{5+x} \times \frac{\sqrt{20-x}+5}{\sqrt{20-x}+5}$ $= \lim_{x \rightarrow -5} \frac{20-x-25}{(5+x)(\sqrt{20-x}+5)}$ $= \lim_{x \rightarrow -5} \frac{-1}{\sqrt{20-x}+5}$ $= -\frac{1}{10}$	1 for multiplying by conjugate 1 correct simplification & answer																			

Solutions to Questions	Marking Scheme	Comments
Question 1 continued		
(d) $(1+z)^8 = \binom{8}{0} + \binom{8}{1}z + \binom{8}{2}z^2 + \dots + \binom{8}{7}z^7 + \binom{8}{8}z^8$ Let $z = 1$ then $2^8 = \binom{8}{0} + \binom{8}{1} + \binom{8}{2} + \dots + \binom{8}{8}$ $\therefore$ sum of coefficients = 2^8 or 256.	1 correct expansion 1 correct substitution of $z = 1$ and the correct answer.	
 (e) (i) Since $3x^2 - y^2 = 12$ then we get $6x - 2y \frac{dy}{dx} = 0 \quad \therefore \frac{dy}{dx} = \frac{3x_1}{y_1} \text{ at } T.$ $\therefore$ equation of tangent: $y - y_1 = \frac{3x_1}{y_1}(x - x_1)$ (ii) since this tangent meets the line $x = 1$ $\therefore y = \frac{3x_1 - 12}{y_1} \text{ \& since } F(4, 0)$ $\therefore FM_{\text{grad}} = -\frac{(x_1 - 4)}{y_1} \text{ \& } FT_{\text{grad}} = \frac{y_1}{x_1 - 4}$ $\therefore FM \times FT = -1 \quad \therefore FM \perp FT.$	1 correct differential 1 gradient at T. 1 correct tangent equation 1 + 1 for correct gradients 1 correct justification of $FM \perp FT$.	

Solutions to Questions	Marking Scheme	Comments
Question 2		
(a)(i) $z = 2 \operatorname{cis} \left(-\frac{\pi}{4} \right)$ (ii) $z^{22} = \left[2 \operatorname{cis} \left(-\frac{\pi}{4} \right) \right]^{22}$ by De Moivre's Theorem $= 2^{22} \left[\cos \left(-\frac{22\pi}{4} \right) + i \sin \left(-\frac{22\pi}{4} \right) \right]$ $= 2^{22} \left[\cos \left(\frac{11\pi}{2} \right) - i \sin \left(\frac{11\pi}{4} \right) \right]$ $= 2^{22} (0 - -i) = 2^{22} i.$	1 for modulus 1 for argument 1 correct use of De Moivre's theorem 1 correct simplification 1 correct answer	
(b)(i) Since $a^2 + b^2 > 2ab$ then similarly $a^2 + c^2 > 2ac$ and $b^2 + c^2 > 2bc$. $\therefore 2(a^2 + b^2 + c^2) > 2(ab + bc + ac)$ $\therefore a^2 + b^2 + c^2 > ab + bc + ac$ as required (ii) Since $a + b + c = 6$ Then $(a + b + c)^2 = 36$ $\therefore a^2 + b^2 + c^2 + 2ab + 2bc + 2ac = 36$ But from part (i) $a^2 + b^2 + c^2 > ab + bc + ac$ $\therefore a^2 + b^2 + c^2 + 2ab + 2bc + 2ac > 3(ab + bc + ac)$ $\therefore 36 > 3(ab + bc + ac)$ $\therefore ab + bc + ac < 12$ as required.	1 correctly writing the other two inequalities. 1 correct inequality 1 correct expansion 1 correct substitution	

Solutions to Questions	Marking Scheme	Comments
Question 2 continued		
(c)(i) $I_n = \frac{1}{n!} \int_0^1 x^n e^{-x} dx, n \geq 0$ By integration by parts $= \frac{1}{n!} \left[-x^n e^{-x} \right]_0^1 - \frac{1}{n!} \int_0^1 [-n x^{n-1} e^{-x}] dx$ $= \frac{1}{n!} [-e^{-1}] + \frac{n}{n!} \int_0^1 [x^{n-1} e^{-x}] dx$ $= -\frac{e^{-1}}{n!} + \frac{1}{(n-1)!} \int_0^1 [x^{n-1} e^{-x}] dx$ $= I_{n-1} - \frac{e^{-1}}{n!} \text{ as required.}$	1 + 1 marks 1 mark	
(ii) $I_4 = I_3 - \frac{e^{-1}}{4!} = I_2 - \frac{e^{-1}}{3!} - \frac{e^{-1}}{4!}$ $= I_1 - \frac{e^{-1}}{2!} - \frac{e^{-1}}{3!} - \frac{e^{-1}}{4!}$ $= I_0 - \frac{e^{-1}}{1!} - \frac{e^{-1}}{2!} - \frac{e^{-1}}{3!} - \frac{e^{-1}}{4!}$ $= \frac{1}{0!} \int_0^1 e^{-x} dx - \frac{e^{-1}}{1!} - \frac{e^{-1}}{2!} - \frac{e^{-1}}{3!} - \frac{e^{-1}}{4!}$ $= -[e^{-x}]_0^1 - e^{-1} \left(\frac{41}{24} \right)$ $= 1 - \frac{65}{24e}$	1 up to this line 1 for this line 1 correct answer	

Solutions to Questions	Marking Scheme	Comments
Question 3  (i) $y = f(x)$	1 shape 1 critical points	

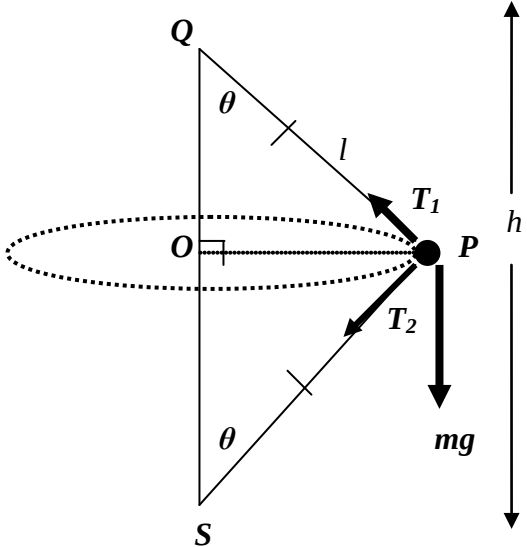
Solutions to Questions	Marking Scheme	Comments
Question 3 continued		
(ii) $y = e^{f(x)}$ 	1 shape 1 asymptote / turning point	
(iii) $y = \ln(f(x))$ 	1 shape 1 x intercept / asymptote	
(b) $(x^2 - 5)(x^2 + 3)$ over rational field $(x - \sqrt{5})(x + \sqrt{5})(x - 3i)(x + \sqrt{3}i)$ over complex field	1 correct factorization 1 correct factorisation	

Solutions to Questions	Marking Scheme	Comments
Question 3 continued		
(c)		
(d) $\int_0^{\frac{\pi}{2}} \frac{1 - \tan x}{1 + \tan x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{\cos x + \sin x} dx$ $= \left[\ln \cos x + \sin x \right]_0^{\frac{\pi}{2}}$ $= 0$	1 correct conversion to sin & cos 1 correct integration + answer	

Solutions to Questions	Marking Scheme	Comments
Question 4		
(a) $T \leq 20 \times 9.8$ But $T = m \omega^2 r$ $\therefore m \omega^2 r \leq 20 \times 9.8$ $\therefore 4 \times \omega^2 \times \frac{1}{2} \leq 20 \times 9.8$ $\therefore \omega^2 \leq 98 \text{ rads/ sec}$ $\therefore \omega \leq \sqrt{98} \times 60 \times \frac{1}{2\pi} \text{ revs / min}$ $\therefore \text{greatest no. of revolutions } 94 \text{ revs/ min}$	1 $T \leq 20 \times 9.8$ 1 1 converting to revs/ min & correct answer	
(b) $\int_a^{a^2} \frac{dx}{x \ln x} = [\ln(\ln x)]_a^{a^2}$ $= \ln(\ln a^2) - \ln(\ln a)$ $= \ln\left(\frac{\ln a^2}{\ln a}\right)$ $= \ln 2$	1 correct integration 1 correct use of log. Rule 1 correct answer	

(c) (i) $\partial V = 2\pi xy \partial x$ $\therefore$ Total volume $= \int_0^{\frac{\pi}{2}} 2\pi x (\cos^2 x - \cos 2x) dx$ $= \int_0^{\frac{\pi}{2}} 2\pi (x - x \cos^2 x) dx$ $= \frac{1}{2} x^2 - \frac{1}{2} x \left(x + \frac{1}{2} \cos 2x \right) \Big _0^{\frac{\pi}{2}}$ $= \frac{\pi^2}{8} - \frac{1}{2} \left(\frac{\pi^2}{8} - \frac{1}{4} - \frac{1}{4} \right)$ $= \left(\frac{\pi^3}{8} + \frac{\pi}{2} \right) \text{units}^3$	1 showing correct volume of slice 1 correct integral + limits 1 correct simplified integral 1 correct use of IBP 1 correct integration by parts 1 correct answer in terms of π.	
(d) $\int_{-4}^4 \cos x (e^x - e^{-x}) dx$ Since $\cos x$ is an EVEN function & $e^x - e^{-x}$ is an ODD function, $\text{EVEN} \times \text{ODD} \rightarrow \text{ODD}$ $\therefore \int_{-4}^4 \cos x (e^x - e^{-x}) dx = 0$	1 + 1 for correct justification 1 correct answer	Note: students do not need to physically find the integral!

Solutions to Questions	Marking Scheme	Comments
Question 5 continued (b)(i) particle moving upwards & acceleration is acting downwards $\therefore a = -\frac{k}{x^2}$ When $x = R$, $a = g$ $\therefore k = gR^2$. $\therefore \frac{d}{dx}\left(\frac{1}{2}v^2\right) = \frac{-gR^2}{x^2}$ since $a = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$ $\therefore \frac{1}{2}v^2 = \frac{gR^2}{x} + c$ Now when $x = R$, $v = u$ initially $\therefore c = \frac{1}{2}u^2 - gR$ $\therefore v^2 = u^2 - 2gR + \frac{2gR^2}{x}$ (ii) when $u = \sqrt{2gR} \rightarrow v^2 = \frac{2gR^2}{x}$ $\therefore v = \frac{dx}{dt} = \sqrt{\frac{2gR^2}{x}}$, $v > 0$ $\therefore \frac{dt}{dx} = \frac{\sqrt{x}}{\sqrt{2gR^2}}$ $\therefore t = \int_R^{4R} \frac{x^{\frac{1}{2}}}{\sqrt{2gR^2}} dx = \frac{1}{\sqrt{2gR^2}} \left[\frac{2}{3} x^{\frac{3}{2}} \right]_R^{4R}$ $\therefore t = \frac{14}{3} \sqrt{\frac{R}{2g}}$	 1 for finding k in terms of g and R . 1 for finding a in terms of g and R 1 integration 1 for c 1 v^2 equation 1 for $\frac{dt}{dx}$ 1 correct integration for t 1 correct answer	

Solutions to Questions	Marking Scheme	Comments
Question 6 (a) 4 players There are 4C_1 ways of choosing the team Then we need four players of the 10: ${}^{10}C_4$ ways The 5 th player is chosen from the remaining 30 in ${}^{30}C_1$ way $\therefore$ total number of ways = ${}^4C_1 \times {}^{10}C_4 \times {}^{30}C_1 = 25200$ 5 players $\rightarrow {}^4C_1 \times {}^{10}C_5 = 1008$ ways $\therefore$ p(at least 4 players from same team) $= \frac{25200 + 1008}{\binom{40}{5}} = \frac{1}{25}$	1 correct answer + 1 justification 1 correct answer 1 correct probability	
(b) 		

Solutions to Questions	Marking Scheme	Comments
Question 6 continued		
(b) (i) By trig. $R = l \sin \theta$ and $h = 2l \cos \theta$. Resolving forces: Vertically: $0 = mg + T_2 \cos \theta - T_1 \cos \theta$. $\therefore mg = (T_1 - T_2) \cos \theta$. $\rightarrow$ ① Horizontally: $m \omega^2 r = T_1 \sin \theta + T_2 \sin \theta$ $\therefore m(l \sin \theta) \omega^2 = (T_1 + T_2) \sin \theta$. $\rightarrow$ ② From ① $\rightarrow T_1 - T_2 = \frac{mg}{\cos \theta} = \frac{2mgl}{h}$ From ② $\rightarrow T_1 + T_2 = ml \omega^2$ ① + ② $\rightarrow T_1 = ml \left(\frac{1}{2} \omega^2 + \frac{g}{h} \right) \rightarrow$ tension in PQ. (ii) ② - ① $\rightarrow T_2 = ml \left(\frac{1}{2} \omega^2 - \frac{g}{h} \right)$ (iii) String PS (hence PQ) will only remain stretched is $T_2 > 0 \therefore \left(\frac{1}{2} \omega^2 - \frac{g}{h} \right) > 0$ $\therefore \omega > \sqrt{\frac{2g}{h}}$ (iv) If $T_1 : T_2 = 2 : 1$ then $\omega^2 = \frac{6g}{h}$ $\therefore \omega = \sqrt{\frac{6g}{h}} \therefore$ period of motion $= \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{6g}{h}}}$	1 $\sin \theta$ and $\cos \theta$ relationships 1 vertical force 1 horizontal force 1 rearrangements of equation 1 correct addition & simplification 1 for answer 1 knowing $T_2 > 0$ 1 showing answer is true 1 getting ω^2 1 for $\omega + 1$ period	

Solutions to Questions	Marking Scheme	Comments
Question 7		
(a)(i) $\alpha^3 + \beta^3 + \gamma^3 = (\alpha + \beta + \gamma)^3 + 3\alpha\beta\gamma$ $\therefore (\alpha + \beta + \gamma)^3 + 3\alpha\beta\gamma$ $= (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \alpha\gamma - \beta\gamma) + 3\alpha\beta\gamma$ $= 0 + 3(-q) = -3q$ (ii) Let $X = \frac{\alpha}{\beta\gamma} = \frac{\alpha^2}{\alpha\beta\gamma} = -\frac{\alpha^2}{q}$ $\therefore \alpha^2 = -qX$ But $\alpha^3 + p\alpha + q = 0$ as α is a root $\therefore -qX\alpha + p\alpha + q = 0$ $(qX - p)\alpha = q$ $(qX - p)^2\alpha^2 = q^2$ $\therefore (qX - p)^2(-qX) = q^2$ $\therefore X(qX - p)^2 = -q$	(1) correct relationship (1) correct expansion (1) correct answer (1) α^2 equation (1) (1) correct solution	
(b) (i) $\therefore$ gradient of normal at P is t^2 $\therefore$ equation is given by $\left(y - \frac{c}{t}\right) = t^2(x - ct)$ Which leads to the required equation. (ii) $N\left(\frac{c - tc^3}{t(t^2 - 1)}, \frac{c - tc^3}{t(t^2 - 1)}\right)$ (iii) $\tan \angle NOP = \frac{\left 1 + \frac{1}{t^2}\right }{\left 1 - \frac{1}{t^2}\right } = \frac{\left 1 + t^2\right }{\left t^2 - 1\right } = \frac{\left 1 + t^2\right }{\left 1 - t^2\right }$ $\tan \angle ONP = \frac{\left 1 + t^2\right }{\left 1 - t^2\right } \therefore \angle NOP = \angle ONP$	(1) correct gradient & use of point – gradient formula. (1) correct coordinates. (1) Finding angles + (1) conclusion $\therefore \triangle ONP$ is isosceles (equal base angles)	Can also show $PN = OP$

Solutions to Questions	Marking Scheme	Comments
Question 8 (a)(i) Since $y = x \tan \theta - \frac{gx^2 \sec^2 \theta}{2V^2}$ ($\div$ by $g \sec^2 \theta / 2V^2$) $\therefore x^2 - \frac{2V^2 \tan \theta}{g \sec^2 \theta} x = \frac{-2V^2}{g \sec^2 \theta} y$ By completing the square: $\therefore x^2 - \frac{2V^2}{g} \sin \theta \cos \theta x + \left(\frac{V^2}{g} \sin \theta \cos \theta \right)^2$ $= \frac{-2V^2 \cos^2 \theta}{g} y + \frac{V^4}{g^2} \sin^2 \theta \cos^2 \theta$ $\therefore \left[x - \frac{V^2}{g} \sin \theta \cos \theta \right]^2 = -\frac{2V^2 \cos^2 \theta}{g} \left[y - \frac{V^2}{2g} \sin^2 \theta \right]$ Which is of the form $(x - h)^2 = 4a(y - k)$. (ii) The focal length is $\frac{1}{4} \times -\frac{2V^2 \cos^2 \theta}{g} = \frac{V^2 \cos \theta}{2g}$ Horizontal range when $y = 0$ $\therefore x \tan \theta - \frac{gx^2 \sec^2 \theta}{2V^2} = 0 \rightarrow x \left(\tan \theta - \frac{gx \sec^2 \theta}{2V^2} \right) = 0$ $\therefore \text{ignore } x = 0, \text{ then } x = \frac{V^2 \sin 2\theta}{g} \therefore \text{since } \frac{2V^2 \cos \theta}{2g} = \frac{V^2 \sin 2\theta}{g}$ we get $\cos \theta = \sin 2\theta = 2 \sin \theta \cos \theta$; $\cos \theta \neq 0$ $\therefore \sin \theta = \frac{1}{2} \rightarrow \theta = \frac{\pi}{3}$	(1) (1) completing the square (1) putting in the form $(x - h)^2 = 4a(y - k)$. (1) focal length (1) for RANGE (1) answer	

Solutions to Questions	Marking Scheme	Comments
Question 8 continued		
(b)(i) $\frac{1}{2}(p+q) \geq \sqrt{pq}$. RTP $\frac{1}{4}(p+q)^2 - pq \geq 0$ LHS = $\frac{1}{4}(p^2 + 2pq + q^2 - 4pq)$ = $\frac{1}{4}(p-q)^2 \geq 0$	(1) mark	
(ii) $\frac{1}{2}(p+q) \geq \sqrt{pq}$. now $\div$ by $\sqrt{q} > 0$ $\rightarrow \frac{1}{2}\left(\frac{p}{\sqrt{q}} + \frac{q}{\sqrt{q}}\right) \geq \sqrt{p}$ $\rightarrow \frac{1}{2}\left(\frac{p}{\sqrt{q}} + \sqrt{q}\right) \geq \sqrt{p}$ as required	(1) mark	
(c)(i) see attached		

James Ruse 2008 MX2 Trial Q8(c) Solution

Q8 (c)(i)

For $n = 1$: This case is trivial.

For $n = 2$:

$$\begin{aligned}\text{LHS} &= \frac{1}{\sqrt{b_2}} B_2 + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 \\ &= \frac{1}{\sqrt{b_2}} (b_1 + b_2) + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) b_1 \\ &= \frac{b_1}{\sqrt{b_2}} + \sqrt{b_2} + \sqrt{b_1} - \frac{b_1}{\sqrt{b_2}} \\ &= \sqrt{b_2} \quad \text{as required. Therefore, true for } n = 2\end{aligned}$$

Assume result holds up to some $n = k$ (strong induction), that is

$$\frac{1}{\sqrt{b_k}} B_k + \left(\frac{1}{\sqrt{b_{k-1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k-1} + \cdots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 = \sum_{r=1}^k \sqrt{b_r}$$

For $n = k + 1$

$$\begin{aligned}\text{LHS} &= \frac{1}{\sqrt{b_{k+1}}} (B_{k+1} + \left(\frac{1}{\sqrt{b_k}} - \frac{1}{\sqrt{b_{k+1}}} \right) B_k + \left(\frac{1}{\sqrt{b_{k-1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k-1} + \cdots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1) \\ &= \frac{1}{\sqrt{b_{k+1}}} (B_{k+1} - B_k) + \left(\frac{1}{\sqrt{b_k}} B_k + \left(\frac{1}{\sqrt{b_{k-1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k-1} + \cdots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 \right) \\ &= \frac{1}{\sqrt{b_{k+1}}} (b_{k+1}) + \sum_{r=1}^k \sqrt{b_r} \\ &= \sqrt{b_{k+1}} + \sum_{r=1}^k \sqrt{b_r} \\ &= \sum_{r=1}^{k+1} \sqrt{b_r}\end{aligned}$$

Therefore, since the initial case and two consecutive cases hold, by the principle of Mathematical Induction, the proposition is true $\square$.

Q8 (c)(ii)

$$\begin{aligned}\sum_{r=1}^n \frac{a_r}{\sqrt{b_r}} &= \frac{1}{\sqrt{b_k}} A_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) A_{n-1} + \cdots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) A_1 \\ &\leq \frac{1}{\sqrt{b_k}} B_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) B_{n-1} + \cdots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 \quad \text{since } A_i \leq B_i \text{ for all } 0 < i \leq n \\ &= \sum_{r=1}^n \sqrt{b_r}\end{aligned}$$

Q8 (c)(iii)

Since all $a_r > 0$ and $b_r > 0$ for all $0 < r \leq n$, therefore from Q8(b)(ii)

$$\sqrt{a_r} < \frac{1}{2} \left(\frac{a_r}{\sqrt{b_r}} + \sqrt{b_r} \right), \quad \text{for all } r \text{ where } 0 < r \leq n$$

Summing over n terms gives:

$$\begin{aligned} \sum_{r=1}^n \sqrt{a_r} &\leq \sum_{r=1}^n \frac{1}{2} \left(\frac{a_r}{\sqrt{b_r}} + \sqrt{b_r} \right) \\ &= \frac{1}{2} \left(\sum_{r=1}^n \frac{a_r}{\sqrt{b_r}} \right) + \frac{1}{2} \left(\sum_{r=1}^n \sqrt{b_r} \right) \quad (\text{rearranging the sum}) \\ &\leq \frac{1}{2} \left(\sum_{r=1}^n \sqrt{b_r} \right) + \frac{1}{2} \left(\sum_{r=1}^n \sqrt{b_r} \right) \quad (\text{using the result from Q8(c)(ii)}) \\ &= \sum_{r=1}^n \sqrt{b_r} \quad \text{as required} \end{aligned}$$